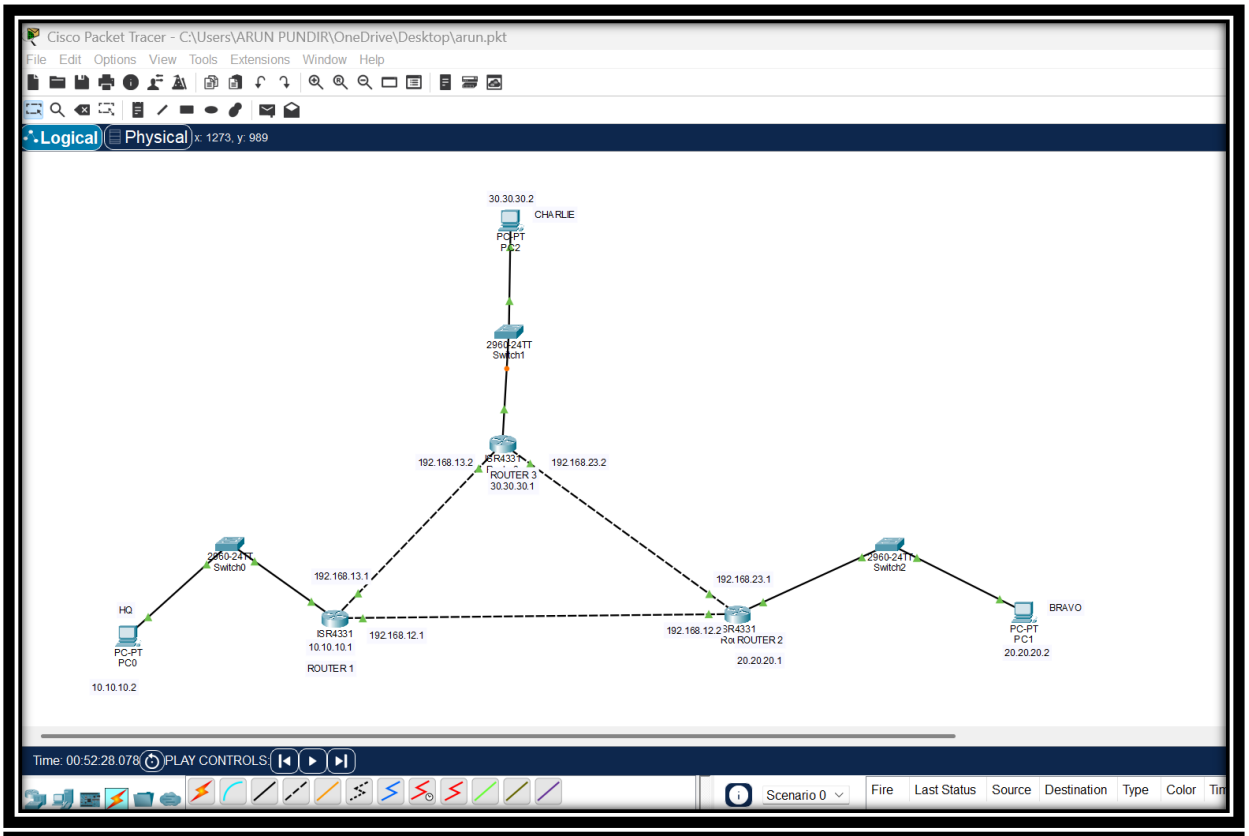
Operation Silent Perimeter – CAPT ARUN PUNDIR

DIAGRAM AND ROUTING TABLES




```
R3#
R3#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     30.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       30.0.0.0/8 is directly connected, GigabitEthernet0/0/2
L       30.30.30.1/32 is directly connected, GigabitEthernet0/0/2
     192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.13.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.13.2/32 is directly connected, GigabitEthernet0/0/0
     192.168.23.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.23.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.23.2/32 is directly connected, GigabitEthernet0/0/1

R3#
```

```
R2#
R2#show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

O    10.0.0.0/8 [110/2] via 192.168.12.1, 00:16:01, GigabitEthernet0/0/2
     20.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       20.0.0.0/8 is directly connected, GigabitEthernet0/0/1
L       20.20.20.1/32 is directly connected, GigabitEthernet0/0/1
     192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/24 is directly connected, GigabitEthernet0/0/2
L       192.168.12.2/32 is directly connected, GigabitEthernet0/0/2
O    192.168.13.0/24 [110/2] via 192.168.12.1, 00:00:15, GigabitEthernet0/0/2
     192.168.23.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.23.0/24 is directly connected, GigabitEthernet0/0/0
L       192.168.23.1/32 is directly connected, GigabitEthernet0/0/0

R2#
```

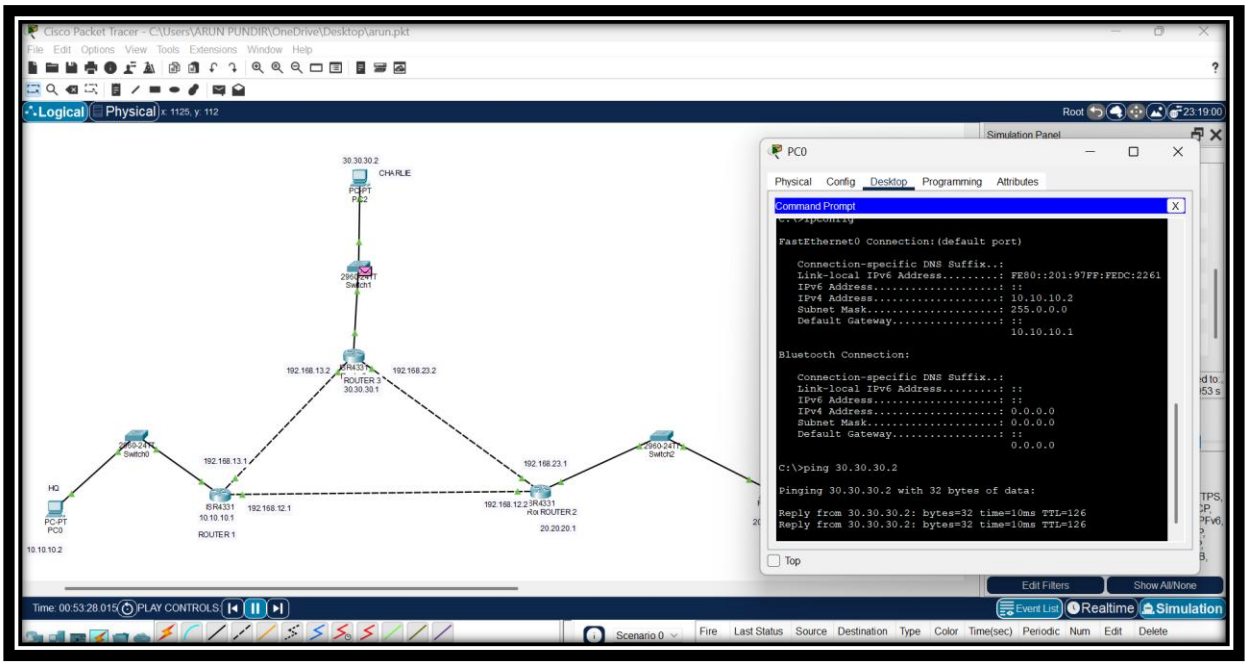
```
R1>show ip route
Codes: L - local, C - connected, S - static, R - RIP, M - mobile, B - BGP
       D - EIGRP, EX - EIGRP external, O - OSPF, IA - OSPF inter area
       N1 - OSPF NSSA external type 1, N2 - OSPF NSSA external type 2
       E1 - OSPF external type 1, E2 - OSPF external type 2, E - EGP
       i - IS-IS, L1 - IS-IS level-1, L2 - IS-IS level-2, ia - IS-IS inter area
       * - candidate default, U - per-user static route, o - ODR
       P - periodic downloaded static route

Gateway of last resort is not set

     10.0.0.0/8 is variably subnetted, 2 subnets, 2 masks
C       10.0.0.0/8 is directly connected, GigabitEthernet0/0/0
L       10.10.10.1/32 is directly connected, GigabitEthernet0/0/0
O    20.0.0.0/8 [110/2] via 192.168.12.2, 00:16:01, GigabitEthernet0/0/2
     192.168.12.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.12.0/24 is directly connected, GigabitEthernet0/0/2
L       192.168.12.1/32 is directly connected, GigabitEthernet0/0/2
     192.168.13.0/24 is variably subnetted, 2 subnets, 2 masks
C       192.168.13.0/24 is directly connected, GigabitEthernet0/0/1
L       192.168.13.1/32 is directly connected, GigabitEthernet0/0/1
O    192.168.23.0/24 [110/2] via 192.168.12.2, 00:00:05, GigabitEthernet0/0/2

R1>
```

SUCCESSFUL PING



DELIBERATE ISSUE

```
Router#config t
Enter configuration commands, one per line.  End with CNTL/Z.
Router(config)#hostname R2
R2(config)#
R2(config)#int gig0/0/1
R2(config-if)#ip add 192.168.23.1 255.255.255.0
R2(config-if)#no sh

R2(config-if)#
R2(config-if)#int gig0/0/2
R2(config-if)#ip add 192.168.12.2 255.255.255.0
R2(config-if)#no sh

R2(config-if)#
R2(config-if)#int gig0/0/1
R2(config-if)#ip add 20.20.20.1 255.0.0.0
R2(config-if)#no sh
R2(config-if)#
R2(config-if)#router ospf 1
R2(config-router)#
```

COMMANDS USE TO IDENTIFY ISSUE

show ip interface brief

show ip route

show running-config

```bash
interface g0/0/1
 ip address 192.168.23.1 255.255.255.0

interface g0/0/2
 ip address 192.168.12.2 255.255.255.0

interface g0/0/1    ❌ CONFIGURED AGAIN
 ip address 20.20.20.1 255.0.0.0
```

show ip ospf neighbor

WHY FAILURE OCCURRED

1. **Wrong Interface IP**
2. **Routing Protocol Failure**
3. **Route Not Advertised**
4. **HQ cannot reach Charlie**
5. **Routing table incomplete**